
4 Collaborative Deep Dive

Master of Science in Engineering
HS 2026

Reader

Feature Engineering im Modul FTP_MachLe

Why This Group Work?

Feature engineering is one of the most *decisive* steps in machine learning. While modern models can be extremely powerful, their success still depends heavily on how the data is represented.

This group work is designed to:

- get you together and help you build your alumni network,
- turn you into *active contributors* rather than passive consumers,
- let you explain concepts in *your own language*,
- produce high-quality documentation that can be used *1:1 as an exam cheat sheet*,
- and, most importantly: have fun while doing serious machine learning.

There are no grades, no competition, and no bashing. The goal is not perfection — the goal is *shared understanding*.

Overall Task

Each group contributes to a joint **blog-style Jupyter notebook**.

Deliverable

- A **ca. 3-page blog post** (Jupyter notebook)
- Clear, compact explanations in simple but precise language
- Minimal, non-trivial examples (prefer clarity over complexity)
- Diagrams, figures, and code snippets where helpful

- A final section: **Takeaways**
- Names of all authors
- Upload to Moodle (Chapter 4)

If all authors agree, the best submissions will be published as blog posts on our GitHub page.

Rules of the Game

- You are all editors, but it makes sense for a volunteer to act as editor-in-chief to orchestrate and compile the different contributions to one consistent document.
- Divide tasks early and merge later (*divide et impera*)
- All sources are allowed (web, books, tools, LLMs)
- Cite sources for figures and diagrams (URLs are sufficient)
- Be respectful, friendly, and inclusive
- Time is short — aim for an *80% solution*

Time Schedule

5'	Introduction and task splitting
40'	Group work on topics
5'	Break
25'	Merge, refine, upload
20'	First 3 presentations
15'	Break
30'	Remaining presentations
15'	Discussion and feedback

Final Section: Takeaways

Every group *must* end their section with a concise list of takeaways:

- What should every ML practitioner remember?
- What are common pitfalls?
- When does this technique really matter?

Closing Remark

Feature engineering is not about tricks — it is about *understanding data*. If you can explain these concepts clearly to your peers, you have truly understood them. Enjoy the process — and enjoy learning from each other.

Group Groups

Each group prepares one clearly structured section in the shared notebook.

Group 1: Feature Normalization

- (i) Why normalization matters
- (ii) Which learners depend on it (kNN, SVM, linear models, neural networks)
- (iii) Standard scaling (z-transform)
- (iv) Min–max scaling
- (v) Robust scaling (median and IQR)
- (vi) Skewness and power transforms:
 - Box–Cox
 - Yeo–Johnson
- (vii) Outlook for neural networks: vanishing and exploding gradients

Group 2: Feature Transformation

- (i) Why and when feature transformations make sense
- (ii) Log, square root, and monotonic transforms
- (iii) Categorical features:
 - Nominal → one-hot encoding
 - Ordinal → label encoding
- (iv) Problems with one-hot encoding (sparsity, curse of dimensionality)
- (v) polynomial features
- (vi) Periodic features:

- Sine / cosine encoding
- Radial basis functions

Group 3: Data Cleaning and Integration

- (i) Missing values and `np.nan`
- (ii) Filtering and smoothing (moving averages)
- (iii) Binning and discretization
- (iv) Missing value imputation:
 - Forward / backward fill
 - Median imputation
- (v) Outlier detection and removal
- (vi) Data integration: merging pandas dataframes

Group 4: Feature Selection I

- (i) Why feature selection helps (bias–variance tradeoff)
- (ii) Univariate feature selection:
 - Pearson correlation
 - F-regression
 - MIC
 - χ^2

Group 5: Feature Selection II

- (i) Regularization as feature selection
- (ii) Lasso regression
- (iii) Model-based feature selection (random forest)
- (iv) Recursive Feature Elimination (RFE)

Group 6: Text Standardization and Encoding I

- (i) Text cleaning using re / regex
- (ii) Word tokenization
- (iii) Normalization:
 - Stemming
 - Lemmatization
 - Difference between both
- (iv) Bag-of-Words model
- (v) TF-IDF

Group 7: Text Standardization and Encoding II

- (i) Historical view: word vectors
- (ii) Word2Vec:
 - Skip-gram
 - CBOW
- (iii) Byte-Pair Encoding (BPE)
- (iv) Modern encodings using LLMs

Group 8: Speech and Audio Data

- (i) Audio feature extraction using librosa
- (ii) Short-Time Fourier Transform (STFT)
- (iii) Mel Frequency Cepstral Coefficients (MFCC)

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